Evaluation of Biochar as an Amendment for the Removal of Metals, Nutrient, and Microplastics in Bioretention Systems

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Abstract: Bioretention systems use plants and soil media to treat pollutants in stormwater. Biochar can be an effective amendment to bioretention soil media for removing pollutants due to its adsorption properties. This study evaluates biochar as an amendment to bioretention soil media for the removal of contaminants in stormwater. Columns with varying ratios of biochar to bioretention soil media (0%, 25%, and 50% biochar) were tested to evaluate the removal efficiencies of copper, zinc, phosphorus, nitrogen, and microplastics from stormwater. Five trials were conducted with stormwater collected from parking lot runoff. Results show that the addition of biochar in bioretention systems improves the removal efficiency of zinc (average of 79% in columns with 50% biochar compared to 54% in columns without biochar) and copper (average of 73% in columns with 50% biochar compared to 25% in columns without biochar). Stormwater effluent from the biochar-amended columns also had significantly lower turbidity (21 NTU) than that of the control columns (58 NTU). Consistent leaching of nitrogen and phosphorus species was evident in the control as well as the biochar-amended columns, with average removal efficiencies of −1,114% for total nitrogen and −703% for total phosphorus. Microplastics were effectively removed by all columns, regardless of biochar amendment, with an average removal of 86%. This study indicates that biochar has the potential to reduce metal concentrations in stormwater when used as an amendment to bioretention systems but has a limited impact on nutrients and microplastics. **DOI: 10.1061/JOEEDU.EEENG-7487.** *© 2024 American Society of Civil Engineers.*

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Introduction

Stormwater runoff originates from rain that flows over impervious surfaces such as rooftops, parking lots, and roads. In urban environments, stormwater can pick up contaminants such as fertilizers, metals (Gunawardena et al. 2013), nutrients (Yang and Lusk 2018), and microplastics (Grbić et al. 2020; Werbowski et al. 2021) that then end up in waterways. The concentrations of these pollutants often reach levels that become harmful to aquatic environments (Walsh 2000), negatively changing stream biotic assemblages (EPA 2022c). According to the EPA, nutrient pollution is one of the most significant threats to environmental health due to its widespread nature and costly mitigation (EPA 2022b). Although nitrogen and phosphorus are necessary for thriving aquatic ecosystems, excess loadings of these nutrients lead to accelerated eutrophication (EPA 2022b). Similarly, excess metal concentrations in waterways can damage aquatic ecosystems (EPA 2022a). In addition, microplastics can decrease growth and cellular processes in aquatic species (Redondo-Hasselerharm et al. 2018; Zhang et al. 2020; Hossain and Olden 2022).

Many older cities that have combined sewer systems, such as Seattle, WA, New York City, NY, and Portland, OR, have implemented passive stormwater treatment in the form of green stormwater infrastructure (GSI) to reduce stormwater volumes and subsequently combined sewer overflows (CSOs). Bioretention is a particularly common GSI that utilizes plants and engineered soil mixes to remove pollutants from stormwater. Bioretention soil mixes (BSM) are typically composed of sand and compost (City of Portland 2020; Washington Dept. of Ecology 2018). Pollutants are removed via physical filtration, sorption to the soil, and plant uptake. Effluent from bioretention systems then infiltrates into the native soil or is conveyed to receiving waters with an underdrain. Stormwater treatment by bioretention systems has been shown to reduce the toxicity of stormwater in aquatic ecosystems (McIntyre et al. 2015).

The efficiency of pollutant removal in bioretention systems varies, however. Some studies have found evidence of significant metal removal (Kohlsmith et al. 2021; Muthanna et al. 2007; Davis et al. 2003), while others have reported the export of copper (Herrera Environmental Consultants 2014; Chahal et al. 2016; Li and Davis 2009). Similarly, bioretention has been found to significantly reduce nitrogen and phosphorus concentrations in some studies (Bratieres et al. 2008; Davis et al. 2006; Henderson et al. 2007; Lucas and Greenway 2008; Passeport et al. 2009), while others have seen export or leaching (Poor et al. 2018; Li and Davis 2014; Herrera Environmental Consultants 2014). The compost

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present in BSM is the source of copper, nitrogen, and phosphorus that is exported out of these systems (Hurley et al. 2017; Mullane et al. 2015; Herrera Environmental Consultants 2015). As a result, some municipalities have replaced the compost with another organic material such as shredded bark, wood fiber mulch, or coconut coir (Howie and Lubliner 2021; New Hampshire Dept. of Environmental Services 2008; Maryland Dept. of the Environment 2009; North Carolina Dept. of Environmental Quality 2017). However, the water-holding capacity of compost is needed in regions that experience hot, dry summers like the Western United States for plant health (US Composting Council 2018). Therefore, many northwest cities still require compost in the BSM (City of Portland 2020; Washington Dept. of Ecology 2018).

Microplastics are defined as plastic pieces that are less than five millimeters long (National Ocean Service 2023) and can be classified as primary or secondary, where primary microplastics are particles designed for commercial use, while secondary microplastics are those that have degraded into smaller pieces over time (National Ocean Service 2023). Microplastics have been found in stormwater, as well as fresh and saltwater ecosystems (Schmidt et al. 2017; Sun et al. 2019; Sutton et al. 2016, 2019; Zhu et al. 2021; Grbić et al. 2020). Removal of microplastics in bioretention systems has been investigated by relatively few studies (Braga Moruzzi et al. 2020; Gilbreath et al. 2019; Lange et al. 2021; Smyth et al. 2021), but the results are promising and suggest removal via settling, filtration, and adsorption (Koutnik et al. 2022). In a study conducted by Smyth et al. (2021), 84% of microplastics in the 106–5,000 μm range were removed in a bioretention cell in Toronto, Canada. An average of 91% of microplastics >125 μm was removed in a bioretention system in San Francisco, CA, with more effective removal of larger particles (Gilbreath et al. 2019). Column studies with three types of geomedia found >99% removal of microplastics between 106 and 5,000 μm (Wolfand et al. 2023). However, additional studies are needed in different regions, particularly in areas with hot, dry summers when BSM pore spaces may become larger as the soil dries out.

To mitigate the leaching from BSM media associated with compost and improve pollutant removal, various amendments have been tested such as biochar, water treatment residuals, iron filings, and fly ash (Poor et al. 2019; Erickson et al. 2007; Kandel et al. 2017; Boehm et al. 2020). Water treatment residuals, iron filings, and fly ash target phosphorus removal, whereas biochar has the potential to increase the removal of other pollutants such as fecal indicator bacteria, metals, and trace organic contaminants. Biochar is made from the pyrolysis of biomass and has been used in many agricultural settings as a soil amendment due to its high water-holding capacity as well as metal and nutrient retention properties (Hasan et al. 2021). Another notable feature of biochar is its carbon sequestration ability (Galinato et al. 2010; Smith 2016). The porous structure and large surface area of biochar provide enhanced adsorption of pollutants by providing an abundance of sites for pollutants to sorb to the biochar (Yin et al. 2017). The adsorption properties of biochar are most significantly influenced by the source biomass material and the pyrolysis conditions (Cao et al. 2013; Meyer et al. 2011; Yao et al. 2012). Biochar has been found to remove fecal indicator bacteria and trace organic contaminants (Kranner et al. 2019; Valenca et al. 2021) and has a high sorption capacity for metals such as zinc and copper (van Hien et al. 2020; Komkiene and Baltrenaite 2016; Xu et al. 2013). Studies have shown that biochar limits the leaching of nutrients from soils (Laird et al. 2010; Yao et al. 2012), whereas others have observed an export of phosphorus (Hale et al. 2013; Novak et al. 2009).

Studies that have evaluated biochar-amended BSMs at the laboratory or mesocosm scale have demonstrated similar results; biochar improves copper and zinc removal but is not as effective for nutrient removal (Rahman et al. 2020; Sang et al. 2019; Boehm et al. 2020). This may be due to the removal mechanisms for nutrients; most nutrient species are typically removed via microbial transformation reactions (e.g., nitrification and denitrification) or plant/microbial uptake (Yang and Lusk 2018). Yao et al. (2012) found that ammonia can be sorbed to biochar, but there was minimal sorption of nitrate and phosphate. Biochar may also improve microplastic removal in bioretention systems by improving physical filtration. Recent studies have shown that biochar-only filters and biochar-amended sand filters can efficiently remove microplastics (Siipola et al. 2020; Wang et al. 2020). However, no studies to date have evaluated microplastic removal in bioretention systems amended with biochar under simulated stormwater conditions. Although many studies have evaluated biochar-amended bioretention systems for metals and nutrient removal, removal efficiency may improve if more biochar is used. Most studies amend the BSM with 10%–20% biochar (Poor and Mohamed 2020; Sang et al. 2019). Larger proportions of biochar in the mix may provide more sorption sites for the sorption of metals and ammonia.

This study evaluated the effectiveness of biochar as an amendment to BSM for the removal of contaminants in stormwater. Five trials were conducted with stormwater collected from a parking lot on columns with BSM only (as a control), 75% BSM and 25% biochar, and 50% BSM and 50% biochar. All columns had a saturated zone in the gravel layer. Influent and effluent for each column were analyzed for copper, zinc, phosphorus, nitrogen, and microplastics.

Materials and Methods

Column Studies

Nine 30.5-cm (12-in.) diameter polyvinyl chloride (PVC) mesocosm columns were used for testing (Fig. 1). Each column included a 15.2-cm (6-in.) ponding zone, 61-cm (24-in.) BSM layer, and a 30.5-cm (12-in.) drainage layer mix of river rock and 1.9-cm (¾-in.) gravel rock. A riser pipe with a control valve 30.5 cm (12 in.) from the base of the mesocosm controlled the saturation level. A slotted pipe at the bottom of the column connected to the riser pipe allowed free drainage of stormwater without clogging. Three different soil mixtures were tested in triplicate:

- three columns with standard City of Portland BSM as the control
- three columns with 25% biochar and 75% BSM
- three columns with 50% biochar and 50% BSM

Proportions for biochar and BSM are volume-to-volume (V/V) ratios. The BSM consisted of a blend of 60%–70% loamy sand and 30%–40% compost per the City of Portland requirements (City of Portland 2020). Prewashed biochar was locally sourced from Stormwater Biochar (Portland, Oregon). Biochar is commonly prewashed with DI water to flush out impurities from the pyrolysis process and dried before use. The biochar feedstock was waste material from a lumber mill that processes Douglas Fir (*Pseudotsuga menziesii*) trees. Pyrolysis occurred at a temperature range of 900°C–1,100°C. Butane activity was 9.8 g/100 g dry biochar, and the surface area correlation was 446 m^2/g dry biochar. Organic carbon content was 86.4% of the total dry mass. Table 1 shows the biochar granule size distribution. The biochar was mixed with the BSM by laying out quantities on a tarp and mixing with a shovel. The columns were placed in a greenhouse to control environmental conditions.

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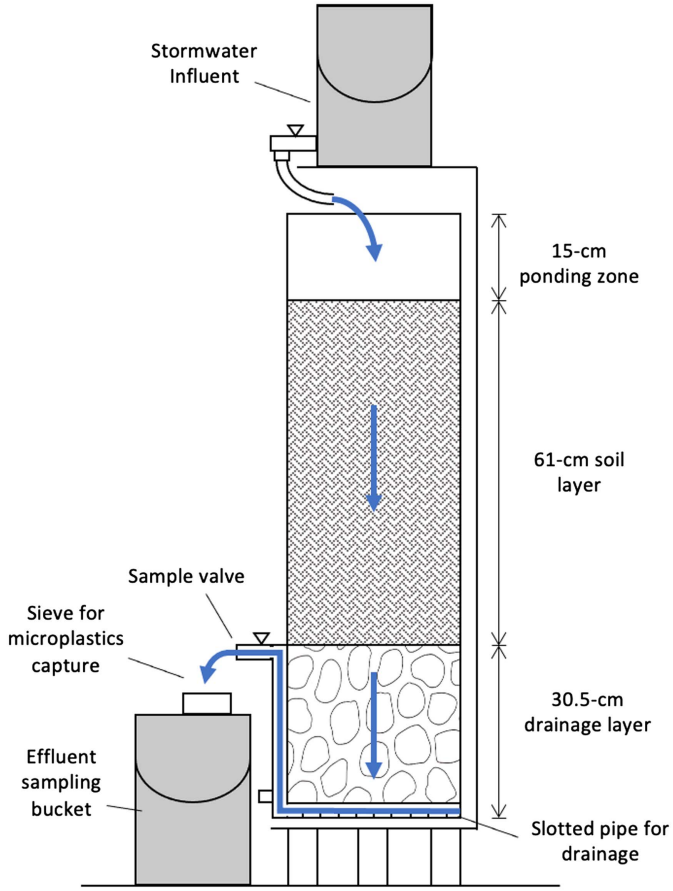

Fig. 1. Experimental setup of columns.

Table 1. Granule size distribution of biochar

Sieve size (mm)	%Passing
<0.5	5.1
1	9.8
2	35.1
4	48.1
8	1.9

Stormwater was collected from a parking lot catch basin on the University of Portland campus using a Dayton portable pump (Model 3YU60A) during three storm events from May to June 2022 and stored in 227-L (60-gal.) high-density polyethylene (HDPE) containers. Before testing, the equivalent of one design storm (24 L or 6.3 gal.) of tap water was applied to all columns to hydrologically compact the soil. Tap water was used instead of deionized (DI) or ultrapure water because tap water has a minimal impact on the chemical composition of the BSM (Stahnke and Poor 2017). Bioretention systems are commonly irrigated with tap water for the first 2 years after installation to help plants become established, so using tap water is representative of field conditions. A total of five trials, or tests, were conducted with approximately one week between each trial to mimic dry periods between rainfall events. During each trial, the stormwater was stirred with a steel rod to resuspend particles and pumped into 25-L polypropylene stormwater containers. Approximately 24 L (6.3 gal.) of stormwater was applied to each column from the 25-L polypropylene containers, which is equivalent to the 6-month 24-h City of Portland design storm (4 cm or 1.6 in.) with a drainage area ratio of 9 and a runoff ratio of 0.9 (City of Portland 2020). Runoff rates were controlled with a spigot and ball valve connected to flexible tubing. The ball valve was continuously adjusted to maintain 5 cm of ponding during testing. It took approximately 10 min for 24 L of stormwater to flow through the columns, which is equivalent to a flow rate of approximately 2.4 L/min. The effluent was collected with a 25-L polypropylene container placed under the sample valve of each column. Microplastics were captured by a 106 μm metal sieve in a wooden sieve holder resting on top of each effluent bucket that was covered when samples were not being collected.

Infiltration rates were measured by weighing the effluent containers before and after each trial and timing the trial. Composite samples were collected from the 25-L polypropylene container using 250-mL HDPE bottles. All containers and sample bottles were cleaned and acid-washed following standard methods (Rice et al. 2012). Turbidity and pH were recorded for both influent and effluent samples.

Leach Tests

Leach tests were conducted on both the BSM and biochar used in the study to determine whether each component is a source of nutrients or metals. Subsamples of the BSM and biochar were set aside before the columns were assembled. The synthetic precipitation leaching procedure (SPLP) EPA Method 1312 (EPA 1994) was followed. Leachate was then analyzed for total phosphorus, total nitrogen, copper, and zinc with the following methods described.

Water Quality Analysis

Each influent and effluent sample was analyzed for total nitrogen (TN), nitrate (NO_3^-), ammonia (NH_3), total phosphorus (TP), and phosphate (PO_4^{3-}) using standard methods (Rice et al. 2012). TN and TP were analyzed using the persulfate method (4500-P and 4500-N, respectively), while NH_3, NO_3^-, and PO_4^{3-} were analyzed using the colorimetric method (4500-NH_3, 4500-NO_3^-, and 4500-P). Reagents used for nutrient analysis were manufactured by Hach (Loveland, Colorado). Metals analysis was conducted using an atomic absorption spectrophotometer (Shimadzu AA-7000, Columbia, Maryland) following standard methods 3500-Cu and 3500-Zn (Rice et al. 2012). Ultrapure water was used for blanks and rinsing (EMD Millipore, Burlington, Massachusetts).

Microplastics were quantified using a wet peroxide oxidation method adapted from Masura et al. (2015). Samples were soaked in 30% H_2O_2 and 0.05 M Fe(II) sulfate solution (VWR, Radnor, Pennsylvania) for 24 h and then heated to 75°C while continuously stirring. Samples were then vacuum-filtered through a prerinsed glass filter apparatus onto 0.45-μm gridded cellulose acetate filters (EZBio, Foxx Life Sciences, Salem, New Hampshire) and dried overnight. Filters were double-rinsed with deionized water before use to minimize potential background contamination. Microplastics were quantified visually using a stereomicroscope (Leica Microsystems Inc. S9i, Buffalo Grove, Illinois), with additional magnification from a 2.0× supplementary lens for magnification up to 110×. A printed grid was used to count all suspected microplastics based on type, shape, and color (Marine and Environmental Research Institute 2015). A photograph of each suspected microplastic was taken using LAS X software and the length and width were measured.

Due to the ubiquitous presence of microplastics in the atmosphere, several quality control measures were necessary during the study. To prevent microplastic contamination of the samples from clothing fibers, bright orange clothing was worn in the laboratory

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and the greenhouse during testing. Any bright orange fibers could then be detected during analysis. Countertops and equipment were wiped down with filtered ethanol solution, and sieves were thoroughly washed with soap and triple rinsed with deionized water. Any laboratory materials, such as jars, filters, and petri slides, were also triple rinsed with deionized water before use. One greenhouse open-air blank was taken to measure the amount of potential background contamination for each trial.

Removal efficiencies for each pollutant were calculated using the following equation:

$$\Delta C = 100 \times \left(\frac{C_{in} - C_{\text{eff}}}{C_{in}}\right) \tag{1}$$

where ΔC = pollutant concentration reduction (percent); C_{in} = influent pollutant concentration (mg/L or particles/L); and C_{eff} = effluent pollutant concentration (mg/L or particles/L).

Statistical Analysis

The average and standard deviation of the three replicates were calculated for each trial. The Kruskal–Wallis and Wilcoxon signed-rank test was used to determine whether there was a significant difference between influent and effluent concentrations, as well as the different treatments (control, 25% biochar, and 50% biochar), and between trials (Helsel and Hirsch 2002). The Kruskal–Wallis test utilizes nonparametric ranking to determine whether there are significant differences between three or more independently sampled groups, and the Wilcoxon signed-rank test compares paired data. The Kruskal–Wallis test was used first to determine whether there was a significant difference between all columns and trials. If a significant difference was found, the Wilcoxon signed-rank test was then used to compare two sets of data at a time. Differences were considered significant if $p < 0.05$. Error bars shown in all figures represent the standard deviation of replicates. All statistical analyses were completed using the statistical package in Excel.

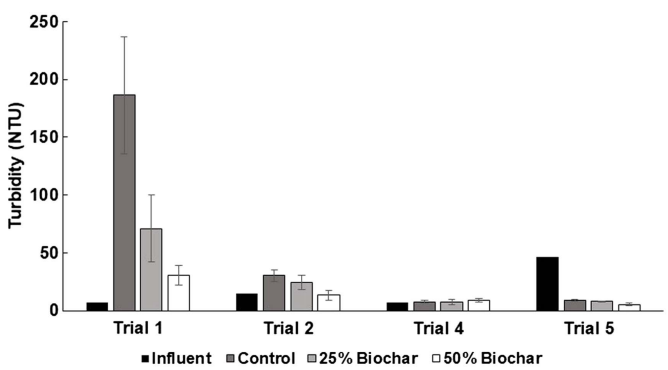

Fig. 2. Average turbidity of the influent, control, 25% biochar, and 50% biochar columns for each trial. Error bars represent the standard deviation between replicates.

Results and Discussion

On average, infiltration rates were found to decrease as the amount of biochar in the mix increased: 176 cm/h (69 in./h) for the control group, 155 cm/h (61 in./h) for the 25% biochar soil mix, and 135 cm/h (53 in./h) for the 50% biochar soil mix columns, across all trials (S1). The average infiltration rate for all columns was 150 cm/h (59 in./h), which is comparable to the average rate of 131 cm/h (51 in./h) observed by Morgan et al. (2020).

Although five experimental trials were conducted, trial 3 was removed from the pH, turbidity, metals, and nutrients analysis due to inconsistencies found with the influent. Metals concentrations in the influent were below detection limits, and TN concentration in the influent was ~10× higher than the other trials. The stormwater may not have been fully mixed before testing, or some other error occurred. Therefore, the results and discussion below only encompass trials 1, 2, 4, and 5.

Turbidity was higher in the effluent than the influent in trials 1, 2, and 4 (Fig. 2), as can be seen by the negative removal efficiencies in Table 2. Turbidity reduction only occurred during trial 5. The negative removal efficiencies are likely due to the flushing of fine particulate matter, which decreased with each trial. Turbidity decreased significantly after trial 1 ($p = 0.004$) but was statistically the same between trials 4 and 5 ($p = 0.82$). Turbidity in the effluent was significantly higher than in the influent during trials 1 ($p = 0.004$) and 2 ($p = 0.04$), but statistically the same during trials 4 ($p = 0.43$) and 5 ($p = 0.50$). Turbidity was generally higher in the control columns, particularly during trial 1, and lower in the columns with 50% biochar, indicating that biochar may improve the removal of small particles that cause turbidity.

The average pH was 7.44 and the median was 7.26. pH remained relatively constant throughout the trials, with no correlation to biochar content. This indicates that biochar has little to no effect on the acidity of water, which is similar to the findings of Hale et al. (2013).

Phosphorus

Effluent concentrations were significantly higher than the influent for both TP ($p = 0.004$–0.009) and phosphate ($p = 0.004$–0.006) during all trials, except for trial 5, indicating export (Table 3, Figs. 3 and 4). Export decreased with each trial ($p = 0.004$–0.02) for both TP and phosphate except between trials 1 and 2 where TP export did not change. Although TP was primarily exported, the removal of phosphate was observed during the last trial (trial 5). This has been observed in other studies where the initial test shows significant export or "flushing" of phosphorus that decreases over time (Poor and Mohamed 2020; Herrera Environmental Consultants 2014). Long-term studies are needed to evaluate at which point phosphorus removal starts to occur.

Differences between treatments were not statistically significant, and all columns exported TP and phosphate. Leach tests indicated the BSM and biochar were sources of phosphorus; TP concentrations were 0.8 mg/L in the BSM and 6.0 mg/L in the biochar. Other studies have found increases in TP and phosphate effluent concentrations with the addition of both biochar (Novak

Table 2. Average turbidity removal efficiencies during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	−2,640	−108	−5	81	−668
25% Biochar	−943	−67	3	83	−231
50% Biochar	−353	8	−18	88	−69

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Table 3. Average phosphorus removal efficiencies during trials

Treatment	Total phosphorus (TP)					Phosphate (PO_4^{3-})				
	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Avg.	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Avg. (%)
Control	−820	−1,195	−265	−105	−596	−1,195	−372	−156	100	−406
25% Biochar	−828	−1,236	−1,044	−246	−839	−1,236	−629	−491	100	−564
50% Biochar	−792	−1,116	−672	−117	−674	−1,116	−351	−275	100	−411

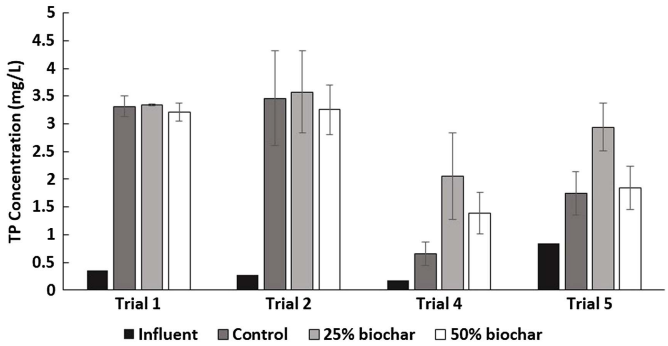

Fig. 3. Average TP concentrations of the influent, control, 25% biochar, and 50% biochar columns for each trial. Error bars represent the standard deviation between replicates.

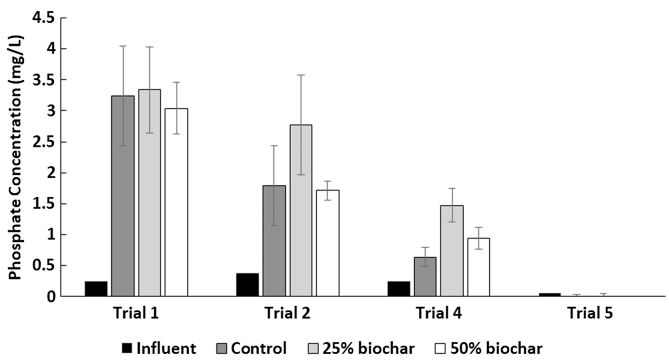

Fig. 4. Average phosphate (PO_4^{3-}) concentrations of the influent, control, 25% biochar, and 50% biochar columns for each trial. Error bars represent the standard deviation between replicates.

et al. 2009; de Rozari et al. 2016) and compost (Hurley et al. 2017; Mullane et al. 2015) to BSM.

Nitrogen

TN was exported from the columns during all trials (S2, Table 4). TN concentrations in the effluent were significantly higher during trials 1 and 2 ($p = 0.004$) but not during trials 4 and 5 ($p = 0.07$–0.31). Export of TN and nitrate is likely from the compost in the BSM and biochar (Novak et al. 2009; de Rozari et al. 2016; Hurley et al. 2017; Mullane et al. 2015; Poor et al. 2018). Leach tests indicated the BSM and biochar were sources of nitrogen; TN concentrations were 2.5 mg/L in the BSM and 23.5 mg/L in the biochar. Although TN concentrations were much higher in the biochar, biochar did not appear to impact the export of TN; TN concentrations were statistically the same between treatments ($p = 0.10$–0.79).

Nitrate removal was apparent only in the columns with 25% and 50% biochar during trial 1 (S3, Table 5). Nitrate concentrations in the effluent were significantly higher during trials 2, 4, and 5 ($p = 0.008$–0.04) but not in trial 1 ($p = 0.19$). Similar to TN, the export of nitrate is likely from the compost in the BSM and biochar. Although less nitrate was leached from the columns with 25% biochar compared to 50% biochar (Table 5), the difference was not statistically significant. Nitrate was statistically the same between treatments ($p = 0.27$–0.85).

Ammonia removal was observed in every trial except trial 1 (S4, Table 6). During trials 4 and 5, almost all ammonia content was removed from the influent—the control, 25% biochar, and 50% biochar treatments removed 99%, 96%, and 76%, respectively, during trial 4. Removal was 100% across all treatments during trial 5 (Table 6). Concentrations of ammonia in the effluent were significantly higher than in the influent during trial 1 ($p = 0.04$) and significantly lower during trials 2, 4, and 5 ($p = 0.006$–0.01). Ammonia concentrations were statistically the same during trials

Table 4. Average TN removal during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	−692	−3,922	−38	−56	−1,177
25% Biochar	−292	−3,883	−74	−257	−1,127
50% Biochar	−171	−3,694	−142	−152	−1,040

Table 5. Average nitrate removal during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	−19	−1,413	−38	−187	−414
25% Biochar	29	−2,127	−56	−123	−569
50% Biochar	68	−2,050	−213	−403	−650

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Table 6. Average ammonia removal during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	−858	23	99	100	−159
25% Biochar	−442	59	96	100	−47
50% Biochar	−400	50	76	100	−44

1 and 2 ($p = 0.21$), but concentrations during trials 4 and 5 were significantly lower than concentrations during trials 1 and 2 ($p = 0.004$–0.009). Differences between treatments were not statistically significant ($p = 0.80$).

Other studies have found that biochar has little to no nitrate sorption ability, while it has been more successful with ammonia (Yao et al. 2012). This is likely due to the controlling removal mechanisms; ammonia is removed via nitrification in the presence of oxygen (transforming ammonia to nitrate) and sorption through cation exchange, whereas TN and nitrate are removed via denitrification in the absence of oxygen. Plant and microbial uptake also reduce nutrient concentrations (Lucas and Greenway 2008), although uptake likely occurs on a longer time scale than the current study (Poor et al. 2018). Despite the saturated zone in the gravel layer, it did not appear to promote enough denitrification to decrease the export. Differences in nitrate and ammonia removal may be due to the microbial community in the BSM as well as the sorption of ammonia; there may have been more nitrifiers than denitrifiers in the soil during testing.

Copper

Copper concentrations in effluent samples were lower than the influent in most columns (Fig. 5) except during trial 2 when copper was exported from the control columns. During trials 4 and 5, copper concentrations in the effluent were below the detection limit in all columns (Table 7). Copper concentrations in the effluent were significantly lower than in the influent during trials 1, 4, and 5 ($p = 0.003$–0.009), but statistically the same during trial 2 ($p = 0.43$). Copper removal was significantly higher in the biochar-amended columns ($p = 0.03$) compared to the control (except in trials 4 and 5 when copper was below detection limits). These results are similar to other studies, which have shown higher removal of copper in biochar-amended BSM compared to the control due to abiotic removal processes such as complexation and cation exchange (Boehm et al. 2020).

Removal was lower in the columns with 25% biochar compared to the columns with 50% biochar (66% compared to 85% in columns with 25% and 50% biochar, respectively) during trial 1, but higher during trial 2 (22% compared to 8% in columns with 25% and 50% biochar, respectively). However, these differences were not statistically significant ($p = 0.69$). The inconsistencies in results may be due to copper leaching from the BSM (Hurley et al. 2017; Mullane et al. 2015). Leach tests indicated the BSM and biochar could be leaching copper; copper concentrations were 44 μg/L in the BSM and 75 μg/L in the biochar. The natural variability of soil and compost may have also caused variable retention of copper in the columns with biochar during trials 1 and 2. Replacing compost completely (and removing the source of copper) with biochar may improve copper removal in bioretention systems, though further studies would be needed to verify.

Zinc

Significant zinc removal ($p = 0.004$–0.009) occurred in all columns during all trials (Fig. 6 and Table 8). Effluent zinc concentrations were significantly lower in the columns with 50% biochar

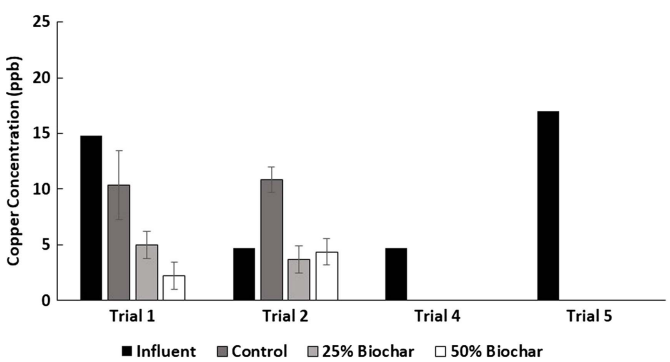

Fig. 5. Average copper (Cu) concentrations of the influent, control, 25% biochar, and 50% biochar columns for each trial. Error bars represent the standard deviation between replicates.

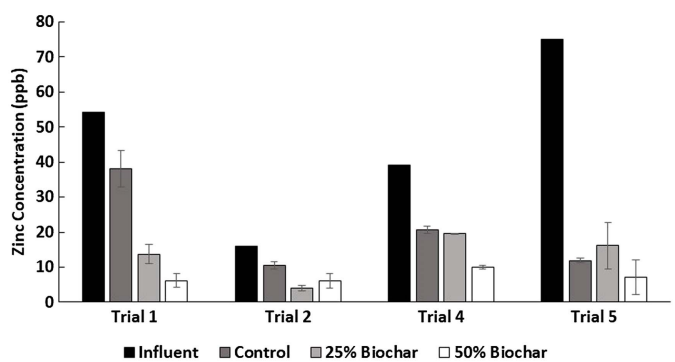

Fig. 6. Average zinc (Zn) concentrations of the influent, control, 25% biochar, and 50% biochar columns for each trial. Error bars represent the standard deviation between replicates.

Table 7. Average copper removal during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	30	−129	100	100	25
25% Biochar	66	22	100	100	72
50% Biochar	85	8	100	100	73

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Table 8. Average zinc removal during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	53	34	47	84	55
25% Biochar	83	75	50	78	72
50% Biochar	89	62	74	90	79

compared to the control ($p = 0.006$) and the columns with 25% biochar ($p = 0.02$) but were statistically the same in the columns with 25% biochar and control columns ($p = 0.30$). The biochar-amended columns generally had higher removal efficiencies, although higher removal was observed from the control columns compared to columns with 25% biochar during trial 5 (Table 8). The columns with 50% biochar had the highest removal efficiency during each trial except trial 2, indicating a dose-response relationship in which more biochar results in greater removal. This is likely due to more available sorption sites with an increase in biochar.

Leach tests indicated the BSM and biochar are likely not sources of zinc; zinc concentrations in both the BSM and biochar were below detection limits. Based on the results of this study, biochar may improve zinc removal if added in high amounts (50% in this study). The range of removal (62%–90%) found in this study is similar to other studies (Xiong et al. 2022; Biswal et al. 2022). Although biochar amounts greater than 50% were not evaluated, completely replacing the compost with biochar may further improve zinc removal. Additional studies would be needed to evaluate a BSM mix with sand and biochar.

Microplastics

Microplastic concentrations were significantly lower ($p = 0.004$) in the effluent compared to the influent (Fig. 7). The average removal rate across all trials and media was 86%, which is within the range of other studies that observed 84%–91% microplastic removal by bioretention systems (Smyth et al. 2021; Gilbreath et al. 2019). Like phosphate and nitrogen, the first trial had significantly higher microplastic concentrations in the effluent compared to the later trials ($p = 0.004$), which corresponds to lower removal (average of 65% versus 93%; Table 9). Biochar did not appear to impact microplastic removal; concentrations of microplastics in the effluent from all columns were statistically the same ($p = 0.19$–0.62). The main removal mechanism of microplastic in bioretention systems is likely physical filtration, and the increased sorption capacity of biochar may not impact removal. Studies have shown that biochar can effectively remove microplastics (Wang et al. 2020; Siipola et al. 2020), but BSM appears to be just as effective.

The most common types of microplastics observed in the effluent were fibers (80%), followed by films (14%) and fragments (6%). The most abundant colors of microplastics found were clear (58%) and blue (24%); however, a wide range of colors were observed throughout the samples. Particle width varied from 7 to 661 μm (median = 17 μm), and length varied from 56 to 6,700 μm (median = 541 μm).

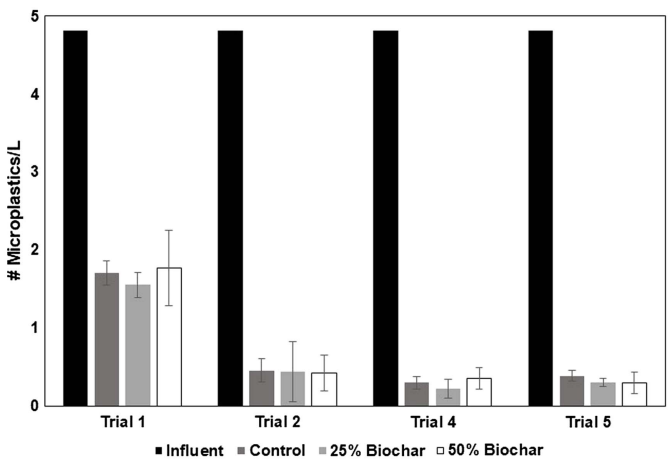

Fig. 7. Average microplastic concentrations of the influent, control, 25% biochar, and 50% biochar columns for each trial. Error bars represent the standard deviation between replicates.

Conclusion

Biochar-amended BSM was evaluated for nutrient, metal, and microplastic removal in a column study. Copper removal was significantly higher in the columns with 25% and 50% biochar than in columns without biochar and the 50% biochar-amended BSM significantly improved zinc removal. The addition of biochar did not impact microplastic removal; all media tested removed microplastics on the order of 91%–95%. Biochar-amended BSM did not appear to impact nitrogen and phosphorus export; export was observed for both the control and biochar-amended columns.

Future research could examine the complete replacement of the compost in bioretention media such that it contains only biochar and sand. Replacement of the compost in BSM with biochar may increase the removal of copper and zinc, as well as eliminate or reduce the export of constituents, such as nutrients, found in compost. In addition, biochar may help plant survival in regions that experience dry summer months due to increased water-holding capacity. Additional work is needed to evaluate the long-term performance of biochar-amended bioretention systems in field settings and with vegetation.

Table 9. Average microplastic removal rate during trials

Treatment	Trial 1 (%)	Trial 2 (%)	Trial 4 (%)	Trial 5 (%)	Average (%)
Control	64	91	94	92	85
25% Biochar	68	91	95	94	87
50% Biochar	63	91	93	94	85

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Data Availability Statement

All data, models, and code generated or used during the study appear in the published article.

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Supplemental Materials

Figs. S1–S4 are available online in the ASCE Library (www.ascelibrary.org).

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